

A Preliminary Phonemic Analysis of Haryanvi

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Abstract

This paper presents a comprehensive preliminary phonemic analysis of Haryanvi (ISO 639-3: bgc), an Indo-Aryan language spoken primarily in the Indian state of Haryana. The present study specifically focuses on the West-Central variety of Haryanvi (WCH), which is natively spoken in parts of the Bhiwani district and the Hansi region of the Hisar district. Based on original lexical data elicited directly from native speakers, minimal pair analysis is utilized to establish and verify phonemic contrasts. The analysis successfully identifies an inventory consisting of thirty-two consonant phonemes, eight vowel phonemes, and three diphthongs. Crucially, this study moves significantly beyond elementary segmental description to investigate broader phonotactic distribution, syllable weight mechanisms within a moraic framework, and highly active phonological processes present in the language. These processes include the positional neutralization of aspirates, stringent retroflex restrictions in word-initial positions, and the application of schwa reduction and syncope. While the dataset utilized in this preliminary study remains limited, the analytical findings contribute substantially to the theoretical understanding of Indo-Aryan phonological typologies and provide a much-needed empirical foundation for WCH, a regional variety that remains critically under-documented within contemporary linguistic literature.

Keywords: Haryanvi, Phonology, Phonemic inventory, Indo-Aryan languages, Syllable weight

1. Introduction

Haryanvi (ISO 639-3: bgc) is a major Indo-Aryan language spoken primarily in the northern Indian state of Haryana and in adjoining regions of Delhi, Rajasthan, and western Uttar Pradesh. Generally classified within the Western Hindi subgroup of the Indo-Aryan language family, it shares several historical and linguistic features with related varieties such as Khariboli and Braj Bhasha (Singh, 1970; Ethnologue, 2021). Despite its widespread use in everyday communication across a vast geographic region, Haryanvi has received relatively limited analytical attention in descriptive linguistic research, particularly concerning its formal phonological architecture and underlying phonological rules.

1.1 Demographic and Sociolinguistic Context

The demographic footprint of Haryanvi is substantial, though precise speaker counts are obscured by Indian census methodologies that typically subsume Haryanvi under the broader category of Hindi for administrative purposes. This subsumption often masks the distinct phonological and syntactic reality of the language. Nonetheless, Ethnologue indicates several million speakers of Haryanvi and its closely related varieties (Ethnologue, 2021). The language if needed is written in the Devanagari script, although its written use remains highly limited compared to its dominance in spoken communication. In formal contexts, standard Hindi is generally favoured, resulting in a diglossic scenario where Haryanvi functions as the basilect and Hindi as the acrolect. Moreover, Haryanvi has a significant diaspora, spoken by communities beyond India, including those that migrated to Pakistan during the 1947 partition of the Indian subcontinent (Madan, 1995).

1.2 Dialectal Variation and the West-Central Variety

Haryanvi is not a monolithic entity; rather, it comprises a complex dialect continuum exhibiting significant phonological and lexical variation across different districts of the state. Among the commonly recognized regional varieties are Bangru (also referred to as Jattu in earlier literature), Deswali, Khadar, Ahirwati, and Bagri (Singh, 1970; Meena and Rao, 2014).

The present study isolates the West-Central variety of Haryanvi (WCH), spoken natively in parts of the Bhiwani district and in the Hansi region of the Hisar district. WCH is strategically selected for this specific study because it occupies a central geographic and linguistic position within Haryana. It functions as a transitional linguistic zone that shares conservative phonetic features with the heavily studied Bangru variety of the north (Singh, 1970) and the Deswali variety spoken in the central and eastern regions. By documenting WCH, this paper aims to provide a baseline phonological model that can be utilized for broader comparative studies across the dialect continuum.

1.3 Research Gap and Objectives

Previous research on Haryanvi has largely focused on grammatical descriptions, prioritizing nominal morphology, verb paradigms, and syntax over the formal acoustic and phonological systems. Earlier descriptive work on related varieties, such as Bangru, provides valuable insights into the grammatical structure of the language but treats phonology largely as an introductory afterthought (Singh, 1959; Singh, 1970). Rigorous theoretical analyses of the phoneme inventory, allophonic variation, syllable weight mechanisms, and phonotactic constraints of Haryanvi remain critically scarce.

To address this critical research gap, the present study asks the following explicit research questions:

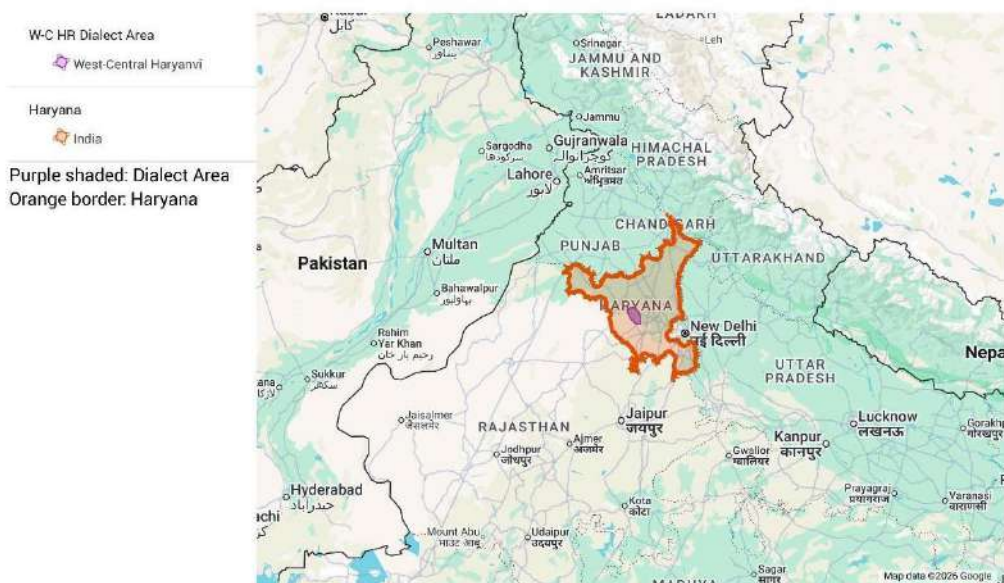
1. What is the precise phonemic inventory of vowels and consonants in WCH, justified through rigorous contrastive distribution and minimal pair analysis?

2. What are the active phonological processes and allophonic variations that dynamically condition these segments in connected speech?
3. What are the syllable structures, moraic weight assignments, and language-specific phonotactic constraints (e.g., sonority sequencing, positional neutralization) governing WCH?

2. Data and Methodology

The data analyzed in this study represent the West-Central variety of Haryanvi natively spoken in parts of the Bhiwani district and in the Hansi region of Haryana. This regional variety belongs to the Western Hindi subgroup of the Indo-Aryan language family (Ethnologue, 2021).

Figure 1. West-Central Haryanvi Variety
(Map created by author using mymaps.google.com)



2.1 Elicitation Protocol and Participant Selection

The primary dataset for this study was collected utilizing the CIIL (Central Institute of Indian Languages) basic vocabulary list, which is designed to elicit culturally and regionally relevant lexemes. The author, functioning as a native speaker of this specific variety of Haryanvi, served as the primary language consultant, providing both elicited forms and intuitive introspective judgments regarding phonotactic grammaticality.

To ensure empirical reliability and guard against idiolectal bias or external interference from standard Hindi, the lexical items, pronunciation norms, and semantic boundaries used in the analysis were rigorously verified through extended consultation with two additional native speakers. Both supplementary consultants originate from the exact same linguistic region (Bhiwani district) and speak WCH as their first language.

2.2 Analytical Framework

The resulting dataset consists primarily of approximately three hundred lexical items utilized to identify phonemic contrasts through minimal pair analysis. Additional lexical material was consulted from available academic glossaries of Haryanvi to cross-check phonological forms and semantic definitions (Jatland Glossary, n.d.).

Minimal pair analysis serves as the foundational, structuralist method for identifying phonemic contrasts in this study. Words exhibiting minimal phonetic differences within identical prosodic environments were meticulously examined to logically establish the phonemic status of both vowels and consonants. Beyond minimal pairs, the methodology incorporates an analysis of complementary distribution to identify allophones and underlying phonological rules.

2.3 Methodological Limitations

As this study represents a preliminary qualitative investigation into the phonological system of WCH, it must be explicitly acknowledged that the current analysis is based on a manually elicited, auditorily transcribed dataset. While auditory transcription by a native speaker provides high categorical accuracy, future quantitative research involving a larger corpus and instrumental acoustic verification utilizing software such as PRAAT will be necessary. Such instrumental methodologies will be critical to precisely map vowel formant frequencies (F1 and F2) and definitively verify Voice Onset Time (VOT) parameters for the robust aspirated plosive series.

3. Phonemic Inventory

3.1 Vowels

The vowel system of the West-Central variety of Haryanvi examined in this study consists of eight monophthongs and three diphthongs. The monophthong inventory includes three short vowels /ə, i, u/ and five long vowels /a:, i:, u:, e:, o:/.

Height	Front (Unrounded)	Central (Unrounded)	Back (Rounded)
Close	/i/, /i:/		/u/, /u:/
Mid	/e:/	/ə/	/o:/
Open		/a:/	

Table 1: Phonemic Vowel Inventory of WCH

As illustrated in Table 1, the WCH vowel system exhibits a pronounced structural asymmetry that is highly characteristic of New Indo-Aryan (NIA) phonologies. While the high vowels and low vowels display a robust, symmetrical quantitative length contrast (short vs. long), the mid vowels /e:/ and /o:/ are exclusively long and lack short

phonemic counterparts within the underlying inventory. This gap is a direct historical reflex of Old Indo-Aryan (OIA) phonology, where original diphthongs /ai/ and /au/ monophthongized into long mid vowels /e:/ and /o:/ in Middle Indo-Aryan, leaving a gap in the short vowel paradigm.

3.1.1 Vowel Contrasts

The phonemic status of these vowels is firmly established through minimal pair analysis, explicitly contrasting short vowels with their phonetically proximate long counterparts.

/kəl/ 'tomorrow/yesterday'	vs.	/ka:l/ 'famine'
/mil/ 'meet'	vs.	/mi:l/ 'mile'
/sun/ 'hear'	vs.	/su:η/ 'swelling'
/lai:/ 'attaching/supporting' (fem.)	vs.	/la:/ 'attach/support' (imperative)

(Note: The verbal root /la:/ in WCH specifically denotes attaching or supporting, contrasting semantically with standard Hindi inferences where a cognate might imply bringing).

3.1.2 Distributional Evidence of Vowels

To demonstrate the full integration of these vowels within the WCH lexicon, their positional distribution across word-initial, medial, and word-final positions is evidenced below. Vowels in WCH exhibit varying degrees of positional restriction.

/ə/: Occurs primarily in initial and medial positions ([ənta:] 'marble balls', [bəhmi:] 'superstitious person').

/a:/: Highly productive across all syllabic positions ([a:a:] 'a low-level shelving unit', [rə[na:] 'to mix up').

/i:/: Found predominantly in initial and medial closed syllables ([ina:m] 'prize', [i:] 'here').

/i:/: Productive across all phonological environments ([i:nt] 'brick', [d̪i:va:] 'lamp', [tʃʰori:] 'young girl').

/e:/: Can appear word-finally, a position strictly prohibited for short vowels ([eq̪di:] 'heel', [li:k̪r̪e] 'small branches').

/u:/: Productive in all structural positions ([ubʰa:ηa:] 'barefooted', [luk] 'hidden').

/u:/: Appears in all structural positions ([u:k̪du:] 'squatting posture', [tu:m] 'heavy silver anklets').

/o:/: Attested word-initially, medially, and finally ([o:t̪ηa:] 'to bear a burden', [mo:q̪q̪a:] 'wandering monk').

3.1.3 Diphthongs

In addition to the monophthong inventory, WCH attests three structurally distinct diphthongs: /əo/, /ai/, and /eu/. In accordance with standard Indo-Aryan metrical phonology, these diphthongs function as heavy, bimoraic nuclei within the syllable structure. Unlike monophthongs, which contrast in length, the diphthongs are inherently heavy and do not contrast with "short" diphthongs.

The positional distribution of these diphthongs is robust across the WCH lexicon, as evidenced by the following elicited items:

/əo/: Occurs primarily in word-initial and medial positions.

- [əo.səŋ.na:] ‘to mix flour with water’
- [bʰrəo.ʈa:] ‘bundle of harvested crops’

/ai/: Attested in both word-initial and word-final positions.

- [ai] ‘(she) came’
- [ɖu.kʰai] ‘to cause pain’

/eu/: Occurs primarily in word-final positions.

- [bə.ʈeu] ‘daughter’s husband’
- [beu] ‘impure, untouchable’

3.2 Consonants

The consonant system of WCH consists of thirty-two phonemes distributed systematically across several places and manners of articulation. The plosive system exhibits a symmetrical four-way contrast based on both laryngeal voicing and aspiration at four distinct places of articulation (bilabial, dental, retroflex, velar). This highly complex laryngeal phonology is a defining typological feature of the Indo-Aryan language family.

3.2.1 Consonant Contrasts

Minimal pairs are provided below to verify the phonemic inventory. These pairs strictly contrast phonetically proximate segments (differing ideally by only one distinctive feature) to isolate the exact acoustic feature responsible for the semantic shift.

Voicing Contrasts (Unaspirated)

/pəl/ (‘moment’)	vs.	/bəl/ (‘strength’)
/ʈa:l/ (‘rhythm’)	vs.	/ɖa:l/ (‘lentil’)
/ʈa:l/ (‘avoidance’)	vs.	/ɖa:l/ (‘branch’)
/ka:l/ (‘famine’)	vs.	/ga:l/ (‘abuse/cheek’)

Aspiration Contrasts

/pəl/ ('moment')	vs.	/p ^h əl/ ('fruit')
/ba:l/ ('hair')	vs.	/b ^h a:l/ ('forehead/search')
/pa:t/ ('door')	vs.	/pa:t ^h / ('lesson')
/la:k/ ('wax')	vs.	/la:k ^h / ('lakh')

Place and Manner Contrasts (Sonorants and Fricatives)

/mən/ ('mind')	vs.	/mən/ ('a unit of weight')
/bəgər/ ('bud powder')	vs.	/bəgər/ ('inner courtyard')
/bəl/ ('force')	vs.	/bəl/ ('burning/twist')
/pa:s/ ('near')	vs.	/pa:f/ ('loop, coil')

3.2.2 Consonant Inventory and Distribution

To analyze the distribution of these phonemes with theoretical clarity, the inventory is separated into Obstruents (Table 2) and Sonorants (Table 3).

Place	Voiceless Unaspirated	Voiced Unaspirated	Voiceless Aspirated	Voiced Aspirated	Fricatives
Bilabial	/p/	/b/	/p ^h /	/b ^h /	(f) ¹
Dental	/t/	/d/	/t ^h /	/d ^h /	/s/
Retroflex	/ʈ/	/ɖ/	/ʈ ^h /	/ɖ ^h /	
Palato-Alv.	/tʃ/	/dʒ/	/tʃ ^h /	/dʒ ^h /	/ʃ/
Velar	/k/	/g/	/k ^h /	/g ^h /	
Glottal					/h/

Table 2: Distribution of WCH Obstruents

Integration of Marginal Phonemes: The labiodental fricative /f/ operates as a marginal phoneme, appearing primarily in borrowed vocabulary (Perso-Arabic or English). In basilectal WCH, this segment is frequently unstable and undergoes rapid integration into the native phonological system by alternating freely with the native bilabial aspirate [p^h]. For example, the word for 'skilled' freely alternates between [fərvət] and [p^hərvət]. Similarly, 'ice' surfaces as both [bəɾəf] and [bəɾəp^h]. This reveals an active phonological filter where foreign fricatives are re-mapped to native aspirated stops.

¹ The 32-count of consonant phonemes represents the *native* phonological inventory, and the marginal/loan phoneme /f/ is excluded from that core count, even though it is mapped in the table.

Place	Nasals	Trill	Flap	Lateral	Approximant
Bilabial	/m/				/v/ / /w/
Alveolar	/n/	/r/		/l/	
Retroflex	/ɳ/		/ɽ/	/ɭ/	
Palatal					/j/

Table 3: Distribution of WCH Sonorants

3.2.3 Distributional Evidence of Consonants

The widespread productivity of the consonantal inventory is evidenced by their unrestricted occurrences in various structural positions within the WCH lexicon, with a few notable exceptions discussed later.

Bilabial plosives ([*panna:*] ‘village block’, [*bangəd*] ‘thick stick’).

Dental plosives ([*tabər*] ‘child’, [*kəd*] ‘when’).

Retroflex plosives ([*rəʰpət*] ‘slap’, [*ha:d*] ‘bone’).

Velar plosives ([*ki[ki:]*] ‘yell’, [*fa:g*] ‘festival’).

Palato-alveolar affricates occur cross-positionally ([*tʃa:m*] ‘lather’, [*gədʒbəŋ*] ‘charming woman’, [*tʃʰe:tʃʰər*] ‘mischief’).

Alveolar sonorants are entirely unrestricted ([*na:[ŋa:]*] ‘to deny’, [*la:mŋi:*] ‘harvesting’).

Retroflex sonorants, however, face strict positional constraints and are barred from word-initial domains.

4. Phonological Processes and Consonant Realization

The phonetic realization of segments in WCH is highly sensitive to word-position and adjacent contiguous segments. Rather than a static one-to-one mapping between underlying phonemes and surface forms, WCH relies on several active, rule-based phonological processes to negotiate articulation.

4.1 Schwa Syncope and Reduction

The short central vowel /ə/ exhibits the most significant allophonic variation in the entire vocalic inventory. In stressed initial syllables, it is fully realized with clear formant structures. However, WCH exhibits a strict Schwa Syncope rule, common in Indo-Aryan languages, where /ə/ is systematically deleted in unstressed medial syllables of multisyllabic words, or word-finally. The formal phonological rule can be generalized as an unstressed schwa deleting in a two-sided open syllable environment. This phonological process frequently results in complex derived consonant clusters at the surface level (e.g., an underlying sequence of /CəCəC/ structurally surfacing as [CCəC] or [CəCC]).

4.2 Positional Neutralization of Aspiration

A defining phonological rule in WCH involves the extensive weakening and positional neutralization of aspiration. While voiceless aspirates (e.g., /p^h/, /k^h/) and voiced aspirates (e.g., /b^h/, /g^h/) contrast robustly in the onset position of a syllable, this vital contrast is neutralized in coda positions. The Voice Onset Time (VOT) lag that distinctly characterizes aspiration is significantly shortened in medial environments, and voiced aspirates are heavily restricted or entirely de-aspirated in word-final codas. This phenomenon serves as a common typological constraint across Indo-Aryan languages designed to preserve respiratory effort at the right edge of a prosodic domain.

4.3 Assimilation and Coarticulation

WCH features highly productive assimilation processes, particularly involving nasality and voicing, to maximize articulatory ease in rapid speech.

Nasal Place Assimilation: The alveolar nasal /n/ frequently undergoes obligatory regressive place assimilation when functioning as a coda before an obstruent onset. It adopts the place of articulation of the following stop to create a homorganic cluster, surfacing as a retroflex [ɳ] before /t/ or /d/, and as a velar [ŋ] before /k/ or /g/.

Voicing Assimilation: Within consonant clusters generated by morphological concatenation, WCH strongly favors regressive voicing assimilation, where the first consonant adopts the [+/- voice] feature of the adjacent trailing consonant to prevent vocal fold interruption.

Contextual Nasalization: Vowels directly preceding nasal codas undergo strong anticipatory nasalization. The velum begins to lower during the production of the vowel. For example, the underlying phoneme /a:/ in the word /ga:m/ ('village') is phonetically realized with a lowered velum as [gã:m].

4.4 Gemination and Compensatory Lengthening

Plosives and sonorants frequently geminate intervocalically ([ed̪d̪i:] 'heel', [gətt̪t̪] 'astonishing', [pan̪n̪a:] 'village block'). Gemination heavily dictates the preceding vowel length, typically demanding a short vowel nucleus within the metrical structure to preserve overall syllable weight limits.

5. Syllable Structure and Moraic Weight

An accurate and comprehensive analysis of the WCH syllable requires moving beyond simple consonant-vowel linear templates and engaging actively with moraic weight and sonority. The syllable serves as the primary domain for stress assignment and phonotactic rules in the language.

5.1 Attested Syllable Templates

WCH permits both complex onsets and codas, generating a maximal syllable template of (C)(C)V(V)(C)(C). The nucleus of the syllable is obligatorily filled by a vowel or diphthong; syllabic consonants are not attested in the data. The dataset demonstrates the following structural patterns:

Structure	Example	Gloss	Syllable Weight
V	/a:/	‘come’	Heavy (2 moras)
CV	/na:/	‘no’	Heavy (2 moras)
CVC	/da:k/	‘jump’	Superheavy (3 moras)
CCV	/pra/	‘prefix in loans’	Heavy (2 moras)
CVC.CCV	/dʱəm.dʱma:/	‘loose structure’	Heavy.Heavy

Table 4: Attested Syllable Structures in WCH

5.2 Syllable Weight and Metrical Phonology

In WCH, syllable weight plays a critical role in stress assignment and morphological processes. Weight is systematically calculated via moras (units of phonological length) (Hyman, 1985), where onset consonants do not contribute to the weight of the syllable:

Light Syllables (1 mora): Consist of a short vowel nucleus with no coda (e.g., (C)V). These are often targeted for reduction or syncope if unstressed.

Heavy Syllables (2 moras): Consist of a long vowel nucleus (C)VV, or a short vowel closed by a single consonant coda (C)VC. These attract primary or secondary stress.

Superheavy Syllables (3 moras): Consist of a long vowel closed by a consonant (C)VVC (e.g., /da:k/), or a short vowel closed by a consonant cluster (C)VCC. Superheavy syllables are heavily restricted and generally only permitted at the right edge (word-finally) of a prosodic domain.

6. Phonotactics and Distributional Asymmetries

Phonotactics refers to the language-specific constraints governing the permissible combinations and positions of phonemes. In WCH, these distributions are driven largely by articulatory preferences and the Sonority Sequencing Principle (SSP).

6.1 Systematic Positional Asymmetries

An analysis of the dataset reveals several distinct phonotactic constraints and striking asymmetries regarding where certain sound classes are permitted to surface:

1. **The Retroflex Sonorant Restriction:** A major distributional asymmetry exists

regarding retroflex sonorants. The retroflex nasal /ŋ/, the retroflex flap /ɭ/, and the retroflex lateral /ɭ/ are strictly restricted to coda and intervocalic positions. They are categorically prohibited from occurring in word-initial onset positions. This structural constraint reflects a robust, pan-Indo-Aryan phonotactic pattern rooted in the historical development of retroflexion from original consonant clusters. (Masica, 1991)

2. **The Glottal Onset Requirement:** Conversely, the glottal fricative /h/ exhibits the exact opposite distributional asymmetry. It is restricted almost exclusively to onset positions. It is strictly prohibited in coda position; even in borrowed lexical items ending in /h/, the segment is either deleted or supported by an epenthetic vowel to forcefully integrate it into an onset role, satisfying the language's acoustic requirement for the glottal fricative to release into a sonorant.
3. **Onset-Preferred Segments:** Aspirated segments, particularly voiceless aspirates like /p^h/ and /k^h/, heavily favor the onset position, aligning structurally with the cross-linguistic tendency to maximize perceptual laryngeal cues at the absolute beginning of a syllable where acoustic salience is highest. (Steriade, 1997)

6.2 Complex Onset Clusters and Sonority

Complex onset clusters (CC) are permitted in WCH but remain highly constrained by universal sonority laws. Permissible native clusters generally adhere strictly to the Sonority Sequencing Principle (SSP), demanding a sharp, uninterrupted rise in sonority from the peripheral edge of the syllable toward the vocalic nucleus (Selkirk 1984; Clements 1990). Consequently, valid native clusters invariably consist of an Obstruent followed by a Liquid or Glide (e.g., /pr/, /kj/).

6.3 Loanword Adaptation and Epenthesis

Clusters violating this required sonority rise (e.g., a liquid followed directly by a stop, or an /s/ + stop cluster like *st-*) are strictly prohibited in the onset. When words containing these illicit sequences are borrowed into WCH from English or standard Hindi, the language repairs the phonotactic violation by systematically triggering vowel epenthesis. An epenthetic vowel, typically /ə/ or /i/, is inserted either prothetically (before the cluster) or anaptyctically (between the consonants) to break up the illicit sequence and conform to the maximal syllable template of the native grammar.

7. Conclusion

This paper has presented a rigorous, data-driven phonological analysis of the West-Central variety of Haryanvi. Advancing well beyond previous introductory morphological descriptions, the study formally establishes an inventory consisting of thirty-two consonant phonemes, eight vowel phonemes, and three diphthongs. Crucially, the analysis analytically reveals that the segmental phonology of WCH is not static, but

is instead heavily governed by dynamic, underlying phonological processes. These processes encompass widespread schwa reduction and syncope, the positional neutralization of laryngeal aspirates in codas, productive regressive assimilation, and strict structural phonotactic constraints against word-initial retroflex sonorants. Furthermore, the syllable structure is demonstrated to be deeply influenced by moraic weight calculation and strict sonority sequencing constraints.

While the reliance on an auditorily transcribed elicited dataset without concurrent instrumental analysis serves as an acknowledged methodological limitation of this preliminary study, the presented findings provide a necessary empirical and theoretical foundation. Future acoustic verification utilizing PRAAT software, alongside comprehensive sociolinguistic research into dialect levelling and language shift within WCH, will be strictly required to further clarify the parameters of allophonic variation. Such ongoing research efforts will be vital to successfully strengthen and formalize the broader documentation of Indo-Aryan linguistic diversity in the twenty-first century.

List of abbreviations

C: Consonant

CIIL: Central Institute of Indian Languages

IPA: International Phonetic Alphabet

OIA: Old Indo-Aryan

MIA: Middle Indo-Aryan

NIA: New Indo-Aryan

SSP: Sonority Sequencing Principle

V: Vowel

VOT: Voice Onset Time

WCH: West-Central Haryanvi

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